

HomeDeck CM4 carrier board
Raspberry Pi CM4 + Touch Display 2 + camera + audio + sensors
All connectivity is by global labels; each sub-sheet is one functional block.

A

power



File: power.kicad_sch

cm4



File: cm4.kicad_sch

usb



File: usb.kicad_sch

video



File: video.kicad_sch

B

sd



File: sd.kicad_sch

audio



File: audio.kicad_sch

sensors



File: sensors.kicad_sch

ui



File: ui.kicad_sch

C

mech



File: mech.kicad_sch

D

E

F

1

2

3

4

5

6

7

8

A

B

C

D

E

F

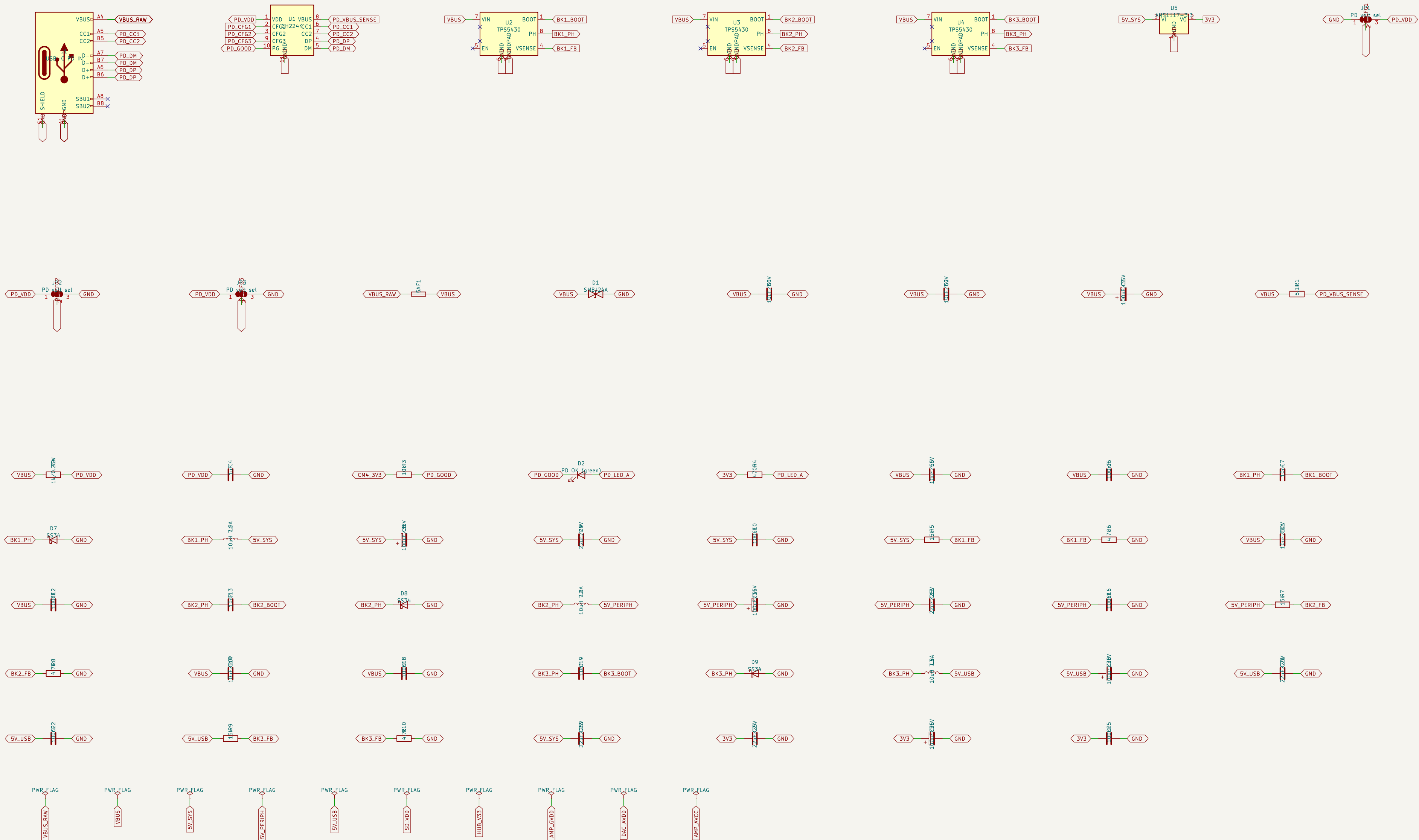
HomeDeck

Sheet: /
File: HomeDeck.kicad_sch

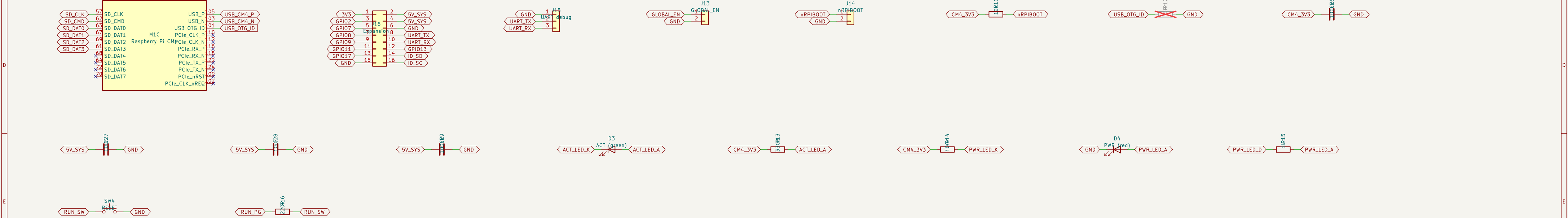
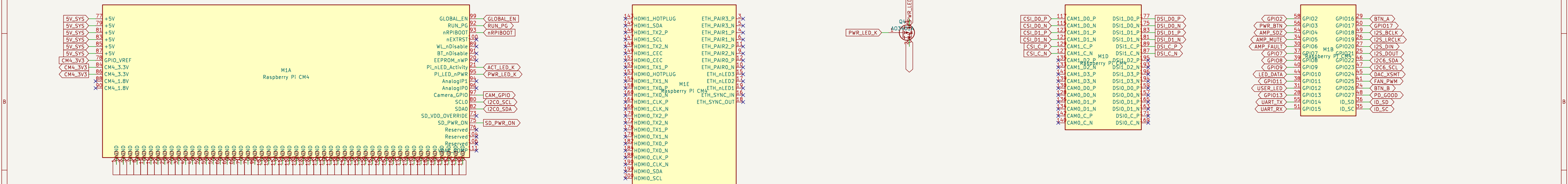
Title: HomeDeck CM4 carrier – smart display / voice assistant

Size: A3	Date: 2026-09-06	Rev: A
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 1/10

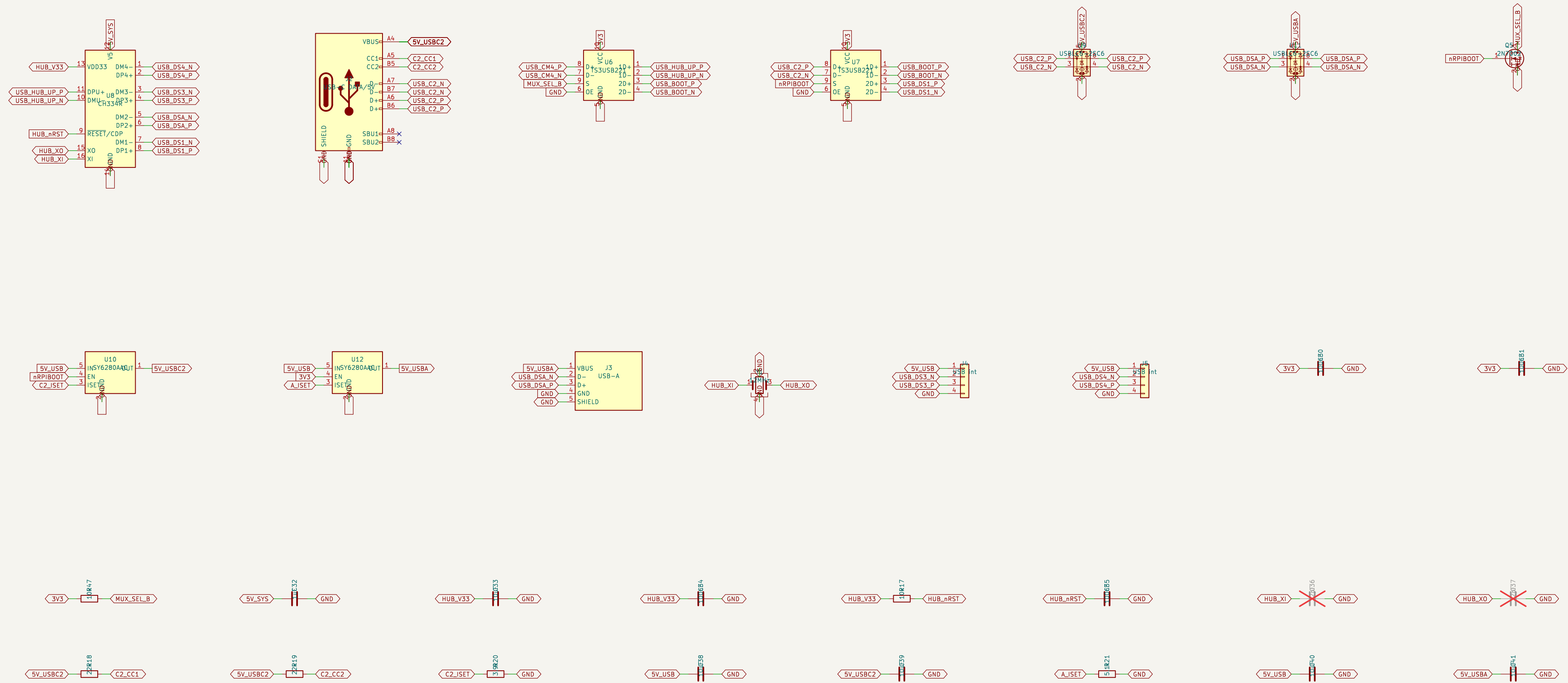
Power: USB-C PD input, 3x TPS5430 bucks, 3.3V LDO



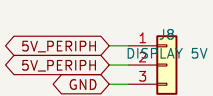
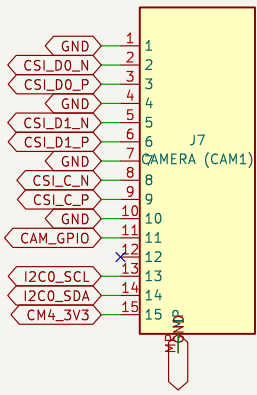
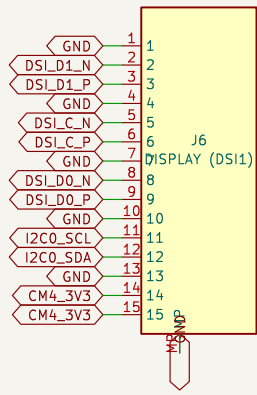
Compute Module 4, LEDs, headers



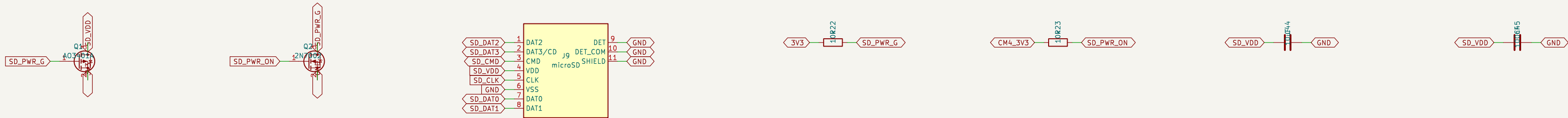
USB: boot mux, CH334R hub, USB-C data port, USB-A, internal headers



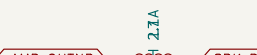
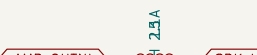
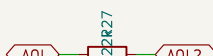
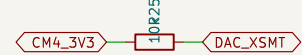
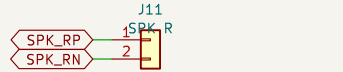
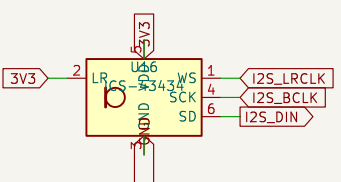
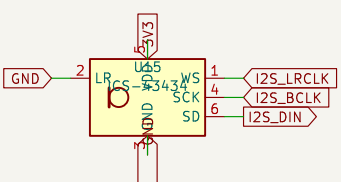
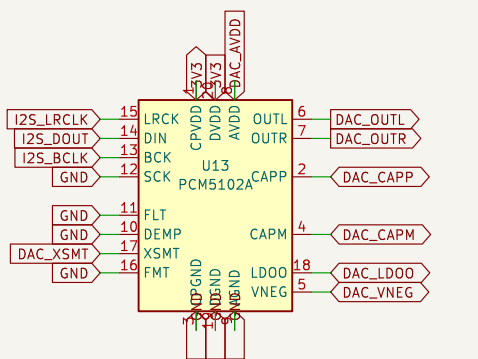
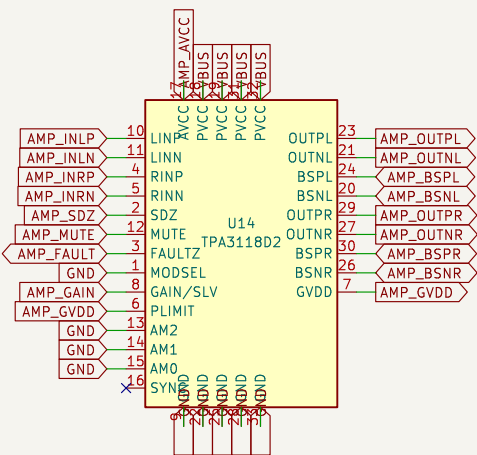
Display (DSI1) and Camera (CAM1) connectors



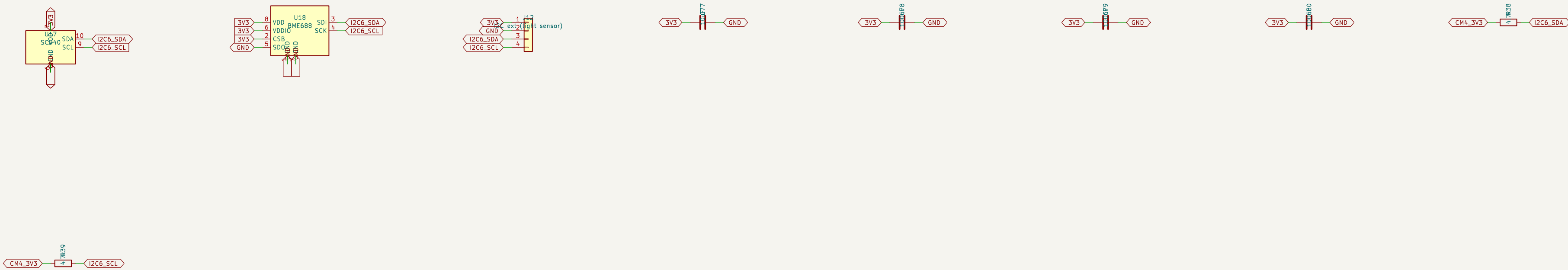
microSD with SD_PWR_ON load switch



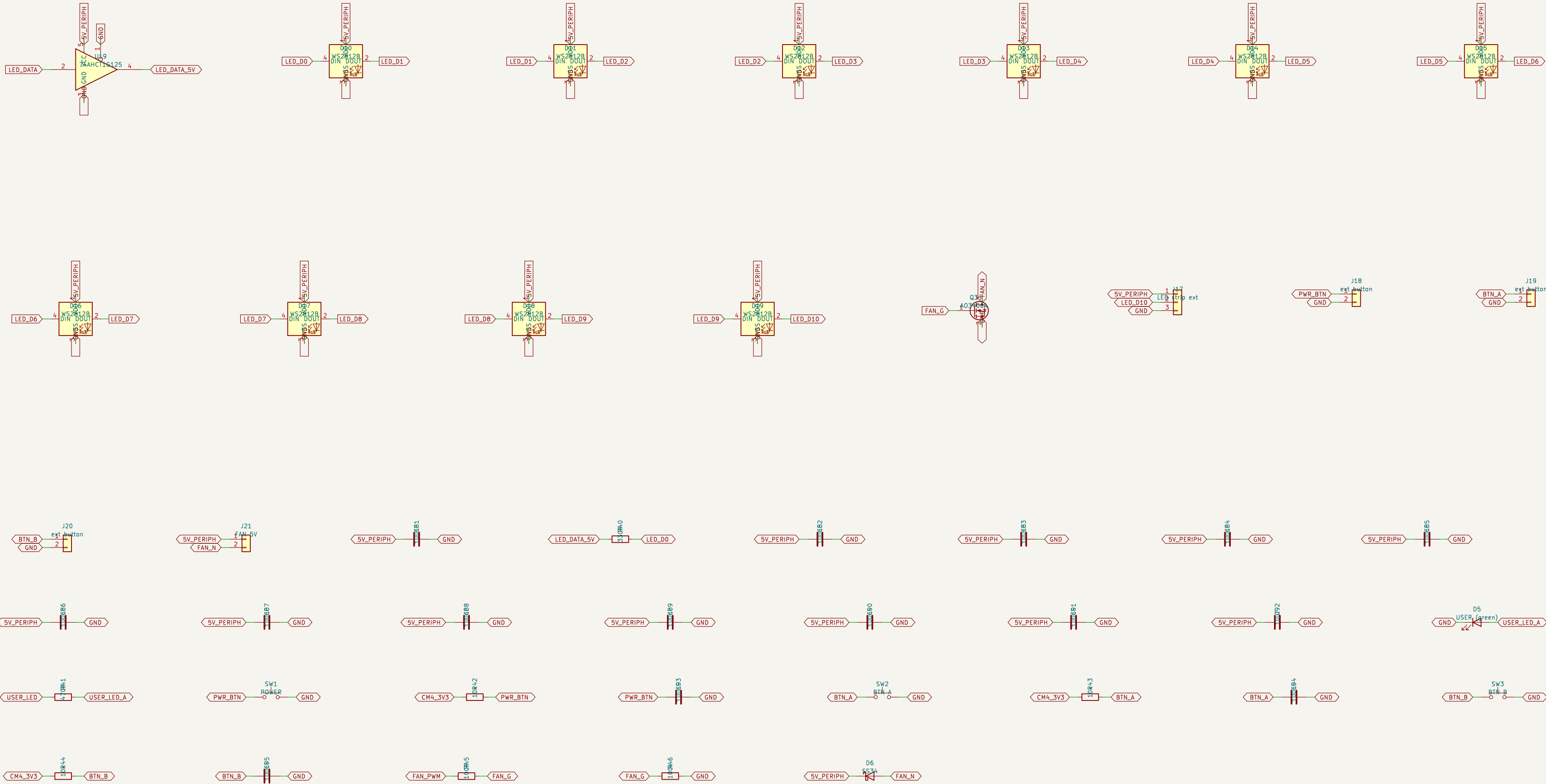
Audio: PCM5102A DAC, TPA3118D2 amplifier, MEMS microphones



Sensors: SCD40 CO2, BME688, I2C header



LED bar, user LED, buttons, fan



Mounting holes

- H1

M2.5 disp. standoff
- H2

M2.5 disp. standoff
- H3

M2.5 disp. standoff
- H4

M2.5 disp. standoff